



ScienceDirect®

Optics & Laser Technology

Date: August 2025

[Show more](#) ▼[Research article](#) [Full text access](#) [Get rights and content](#) ↗[Full length article](#)

Laser transmission welding of transparent polymers based on femtosecond laser surface pretreatment

Dongyang Wang ^{a, b}, Kaige Liu ^c, Jianpang Zhai ^c, Shuangchen Ruan ^d, Yan Kang ^{b, e}  , Qingyang Wu ^a   [Show more](#) ▼[View PDF](#)[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.optlastec.2025.112669](#) ↗[Article](#)[Recommended articles](#)[Cited by \(6\)](#)[Metrics](#)

Abstract

Laser transmission welding (LTW) technology has garnered widespread attention in polymer welding due to its unique advantages. Among the challenges in LTW research of polymers, achieving welds between two transparent polymers without using absorbers is one of the most difficult issues. To date, there have been no reports of effective welding of transparent polymers using near-infrared (NIR) lasers without the use of absorbers, because transparent polymers have low absorption rates for commonly used NIR lasers, resulting in insufficient absorption of laser energy to achieve effective welding. This study presents a femtosecond laser surface pretreatment approach that modifies polymer

Outline

[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experimental designs](#)[3. Results and discussion](#)[4. Conclusions](#)[CRediT authorship contribution statement](#)

surface roughness to enhance absorption rates. The effects of various ablation parameters on the roughness and laser absorption of polycarbonate (PC) samples were investigated. Effective welding of PC/PVC was achieved using both NIR (808nm) and mid-infrared (1940nm) lasers, and the shear strength of the welded joints was tested. The applicability of this approach was further validated across multiple transparent polymer sets. By enabling NIR laser transmission welding without absorbers, this method offers significant potential for applications in the biomedical, food, and medical industries.

Keywords

Laser transmission welding; Laser surface pretreatment; Polymer; Bonding strength

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

Polymer materials have wide applications in electronics, healthcare, automotive, and aerospace industries due to their excellent fatigue resistance, toughness, high specific strength, and good biocompatibility [1], [2]. However, as the variety of plastic products increases, the need for secondary joining becomes inevitable. Secondly, joining polymers allows for the creation of products with higher structural complexity. Additionally, composite structures made of different polymers can maximize the unique properties of each material, offering even greater application advantages [3]. There are three main methods for effective polymer joining: mechanical joining, adhesive bonding, and welding [4], [5], [6]. Among these, laser transmission welding (LTW) has gained significant attention due to its advantages, including being environmentally friendly, high power, flexible, offering broad spatial selectivity, and non-contact operation [7]. In LTW of polymers, the two materials typically need to exhibit different optical properties: one polymer should be laser-transparent or semi-transparent, while the other should be

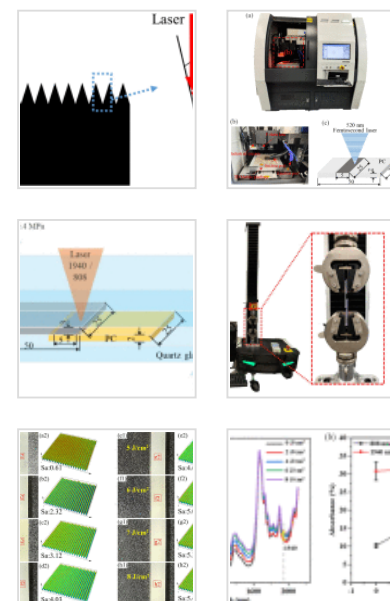
[Declaration of competing interest](#)

[Data availability](#)

[References](#)

[Show full outline](#) 

Figures (19)



[Show 13 more figures](#) 

Tables (1)

 Table 1. Physical properties of PC and PVC.

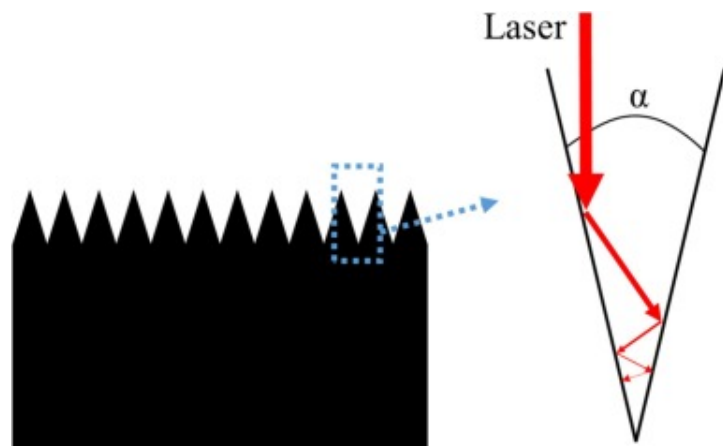
laser-absorbing (opaque) [8]. However, in practical applications, various material combinations exist, with the difficulty of joining generally ranked as follows: transparent/transparent > colored/colored > colored/black [9]. Transparent polymers have weak absorption of commonly used industrial NIR lasers (wavelength < 1200 nm), preventing the absorption of sufficient laser energy to melt the joint and achieve effective welding. As a result, welding two transparent polymers is one of the most challenging tasks in the field of LTW.

To address this challenge, researchers have proposed surface pre-treatment methods such as coating, plating, or simply placing absorbers on the polymer surface [10]. Carbon black was one of the earliest absorbers used due to its cost-effectiveness and strong absorption properties; extensive studies have been reported on its applications [11], [12]. However, the addition of carbon black significantly affects the appearance of the weld and makes it difficult to precisely control the deposition area. In 2002, Gentex commercialized Clearweld®, an infrared absorber invented by the Welding Institute of the UK. The main advantage of Clearweld® is its transparency to visible light and strong absorption in the NIR range, with peak absorption between 0.940 μm and 1.064 μm, attracting considerable research attention since its inception. In 2003, Hilton et al. from the University of Cambridge used a brush to apply Clearweld® to the interface between transparent polymer bottles (PP and ABS) and their lids, and successfully welded the bottles using a linear seam welding method with a semiconductor laser. They also welded two overlapping transparent PMMA plates, each 3 mm thick, by applying Clearweld® to the interface, achieving a weld strength equivalent to 20%–40% of the base material strength [13]. In 2015, Jiang et al. from Heriot-Watt University studied the welding of polymer-based microfluidic devices. To achieve effective welding between two transparent PMMA plates, they proposed using a vacuum electron beam coating system to deposit metal films on the polymer surface as absorbers. They deposited 0.5 μm and 1 μm thick titanium films on 1 mm thick PMMA substrates using electron beam deposition and then welded the substrates with a 970 nm semiconductor laser. By optimizing the processing parameters, they achieved a maximum tensile strength of approximately 6 MPa for the

weld joint, and produced microfluidic welded components with excellent weld quality and no leakage [14]. In 2018, Liu Minqiu et al. from Shenzhen University proposed placing metal wires on the surface of transparent polymers as absorbers before welding. They used metal wires (99.99% purity, 0.2 mm diameter) made of Fe and Cu, placed on the interface between two 1.5 mm thick transparent PET plates, and successfully performed transmission welding using an 808 nm diode laser [15]. They also compared the welding quality between PET using carbon black and metal wires as absorbers. The results showed that metal absorbers, due to their stable physical and chemical properties, provided higher and more stable weld strength than carbon black, with cleaner and more transparent weld joints free from impurities. In the same year, they solved the issue of excessive gaps between the two PET plates caused by the metal wires as absorbers by milling grooves (0.5 mm wide, 0.3 mm deep) in the PET plates using a micro-ball end mill (0.5 mm diameter) [16].

The aforementioned surface pre-treatment methods enable LTW of two transparent polymers, but they essentially involve adding new materials such as carbon black, Clearweld®, metals, and their oxides as absorbers [7], [8], [17], [18], [19], [20], [21]. Numerous studies have been reported on welding absorbers, and the exploration of new absorbers continues. However, the addition of absorbers often affects the appearance of the joint [22], [23], increases costs, and some absorbers are highly toxic [24], [25], limiting the range of applications for the products. In addition to using absorbers, researchers have proposed using mid-infrared lasers, to which polymers exhibit higher absorption rates, as the welding light source. In this study, a mid-infrared laser welding system with a wavelength of 1940 nm was developed to verify the feasibility of this approach. However, due to the relatively low power and higher cost of mid-infrared lasers, their industrial applications are limited. Furthermore, there has been no reported case of effective welding of transparent polymers using NIR lasers without the use of absorbers. To tackle this challenge, the present study proposes using femtosecond laser ablation to modify the surface of transparent polymers and increase their absorption rate, thereby achieving effective welding between two transparent polymers using a NIR laser

(808nm) without the need for absorbers. The welding strength was found to be comparable to that achieved with the 1940nm mid-infrared laser under similar conditions. By achieving effective welding using NIR lasers without the need for absorbers, this method reduces the potential contamination and toxicity associated with absorbers, lowers the cost of laser welding equipment, and enhances the safety, environmental friendliness, and cost-effectiveness of the technology. Consequently, this approach holds significant potential for applications in the biomedical, food, and medical fields. Specifically, this paper proposes a solution of femtosecond laser ablation of the polymer surface to modify its roughness and thereby increase its absorption rate, as illustrated in [Fig. 1](#). The absorption rate of non-metallic materials is related to the incident angle of the laser, and surface roughness alters this angle, significantly affecting the absorption rate of the material [26]. The mechanism by which surface roughness affects laser absorption can be attributed to two factors. First, when the laser irradiates the material surface, the uneven surface causes multiple reflections, leading to repeated absorption of the laser by the material. Second, the absorption rate of some non-metallic materials changes with the laser's incident angle, reaching a maximum at the Brewster angle. Based on this theory, this study first used femtosecond laser ablation to modify the surface roughness of PC samples and measured the absorption rate of the modified samples. Subsequently, transmission welding of PC/PVC samples was performed using an 808nm NIR laser and a 1940nm mid-infrared laser. The shear strength of the welded joints was evaluated for both laser wavelengths, and their failure mechanisms were analyzed. Additionally, multiphysics simulations of the two welding processes were conducted using COMSOL.



[Download: Download high-res image \(58KB\)](#)

[Download: Download full-size image](#)

Fig. 1. Rough surface causing multiple reflections of laser.

2. Experimental designs

2.1. Experimental materials

The welding materials used in the experiment were PC and Polyvinyl chloride (PVC) sheets, with dimensions of $50 \times 25 \times 2 \text{ mm}^3$ and $50 \times 25 \times 3 \text{ mm}^3$, respectively. Both materials were produced via injection molding. The physical properties of the materials are detailed in [Table 1](#). To remove surface impurities and oil, the sheets were wiped with a 99.99% pure ethanol solution. Additionally, to prevent the influence of moisture from the air on the material properties, the sheets were placed in a drying oven 12h before the femtosecond laser modification experiments.

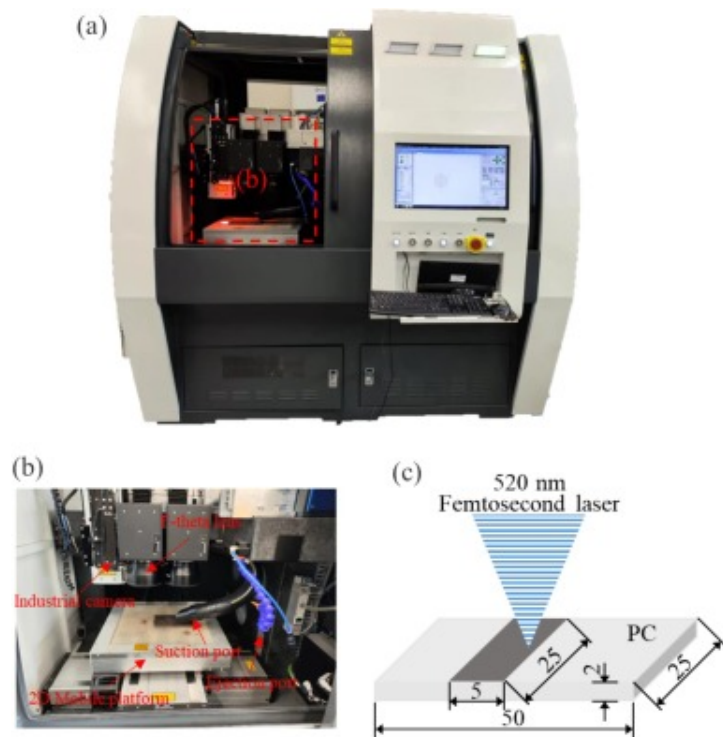
Table 1. Physical properties of PC and PVC.

Materials	Density (g/cm ³)	Tensile strength (MPa)	Transmittance		
			520nm	808nm	1940nm
PC	1.20	54.1	88.3%	90.0%	68.6%
PVC	1.40	44.3	81.9%	89.8%	64.0%

2.2. Experimental equipment

2.2.1. Femtosecond laser ablation modification

The surface modification experiment on PC samples was carried out using a Green Femtosecond Laser (GFL) processing platform with a wavelength of 520nm. The platform is equipped with a high-precision galvanometer system to control the laser scanning path, with a maximum scanning speed of up to 12,000mm/s. [Fig. 2\(a\)](#) and [Fig. 2\(b\)](#) show the photographs of the femtosecond laser processing platform. The maximum average power of the femtosecond laser is 8W, with a repetition frequency of 250kHz and a pulse width of 300fs. The laser beam's focused diameter is approximately 20μm. During the modification experiment, the laser was focused on the surface of the PC sample, and the processing speed was set at 2500mm/s. The laser performed line scanning every 30μm over the processing area, with six repeated scans. Upon completion, an ablation area of 25×5 mm² was formed on the surface of the PC sheet, as shown in [Fig. 2\(c\)](#). To prevent the interference of debris generated during femtosecond laser modification, a suction pipe was placed beside the modification area of the PC sample to absorb debris in real-time.



Download: [Download high-res image \(209KB\)](#)

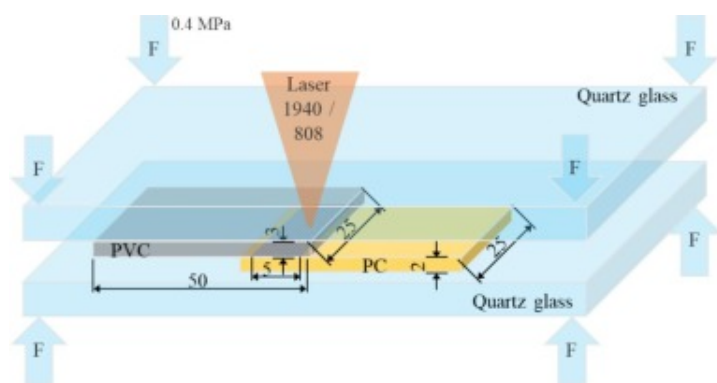
Download: [Download full-size image](#)

Fig. 2. Femtosecond laser processing platform and ablation modification schematic: (a) Femtosecond laser processing platform; (b) Enlarged view of the processing platform; (c) Schematic of laser ablation on a PC sample.

2.2.2. Laser transmission welding

Continuous laser transmission welding was performed on polymers using lasers with wavelengths of 808nm and 1940nm, respectively. The light source for the 808nm welding machine was a semiconductor continuous laser with a maximum output power of 44W. Prior to welding, the PC and PVC samples were overlapped after femtosecond laser ablation ([Fig. 3](#)) and then placed in a fixture. A quartz glass plate with high transmittance for the 808nm laser was placed on the top surface of the PC sample, and the PVC sample

was positioned on top of the same quartz glass. Quartz glass was used as a fixture during the welding process due to its high transmittance and low absorption in the NIR and mid-infrared wavelength ranges, as well as its high thermal conductivity. These properties minimize interaction with the laser beam, avoiding significant energy loss or internal damage to the fixture. The fixture was equipped with a pair of 25mm diameter guide rod cylinders, which served to lift the quartz glass beneath the sample to apply the necessary pressure during welding. Based on the relevant literature and experimental experience regarding pressure studies, the cylinder pressure for the guide rod was set to 0.4MPa [27], [28]. During the welding process, the laser focal point remained fixed, and welding was achieved by moving the two-dimensional platform under computer control. The 1940nm welding machine used was a custom-built mid-infrared laser welding machine. Prior to welding, the PC and PVC samples were also overlapped in the same manner (Fig. 3), with the cylinder pressure similarly set at 0.4MPa. It should be noted that, although effective welding of PC and PVC without ablation modification can be achieved directly using the 1940nm mid-infrared laser, to minimize experimental variables, all PC samples used in this section had undergone femtosecond laser ablation modification. During the welding process, all samples were welded using a single linear contour weld method, with both lasers set to a power of 11 W. The welding strength at different welding speeds was examined.



Download: [Download high-res image \(36KB\)](#)

Download: [Download full-size image](#)

Fig. 3. Laser welding.

2.2.3. Mechanical performance testing and characterization

After welding, the welded joints were mounted in shear fixtures and tested for shear strength using a universal testing machine (model: DECCA-5000, as shown in Fig. 4). To ensure uniform force distribution, 2mm and 3mm rubber pads were added at the sample clamping positions, as shown in Fig. 4(b). The loading rate of the equipment was set to 2mm/min. After the welded samples fractured, the maximum shear force for each sample was recorded, and the shear strength was calculated using the following equation:

$$\sigma = F / (L * B) \quad (1)$$

where σ is the shear tensile strength (MPa), F is the shear tensile force (N), L is the welding length (20mm, fixed value), and B is the welding width (mm). Additionally, to investigate the effects of laser modification and the failure mechanism of dissimilar material joints, a laser scanning confocal microscope (LSCM) was used to examine the overall surface microstructure and roughness of the modified surfaces.



Download: [Download high-res image \(180KB\)](#)

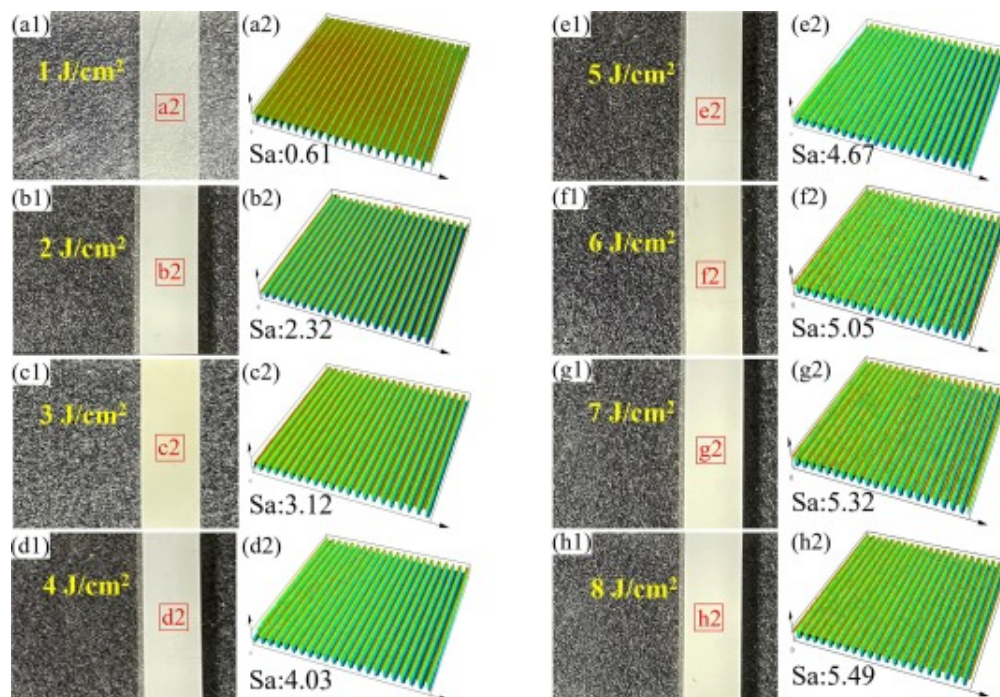
Download: [Download full-size image](#)

Fig. 4. Universal testing machine and shear test schematic: (a) Universal testing machine and enlarged view; (b) Schematic of the shear test.

3. Results and discussion

3.1. Femtosecond laser ablation of PC

To investigate the effect of femtosecond laser energy density on ablation modification and to determine appropriate modification parameters, PC samples were subjected to surface modification experiments using different laser energy densities. Fig. 5 shows the physical appearance and LSCM images of the PC samples after modification with different energy densities. It can be observed that the modified areas exhibit regular shapes consistent with the theoretical design values ($25 \times 5 \text{ mm}^2$). Within the investigated parameter range, the roughness of the modified areas increased with energy density, reaching a maximum at 8 J/cm^2 (S_a : $5.49 \mu\text{m}$). Although S_a serves as a representative parameter for surface roughness to describe the effect of femtosecond laser surface modification, it should be noted that other parameters (such as peak-to-valley ratio, skewness, or texture isotropy) may also slightly influence the absorption rate. This highlights an important direction for future research, which is to investigate the impact of absorption mechanisms through multi-parameter surface characterization.



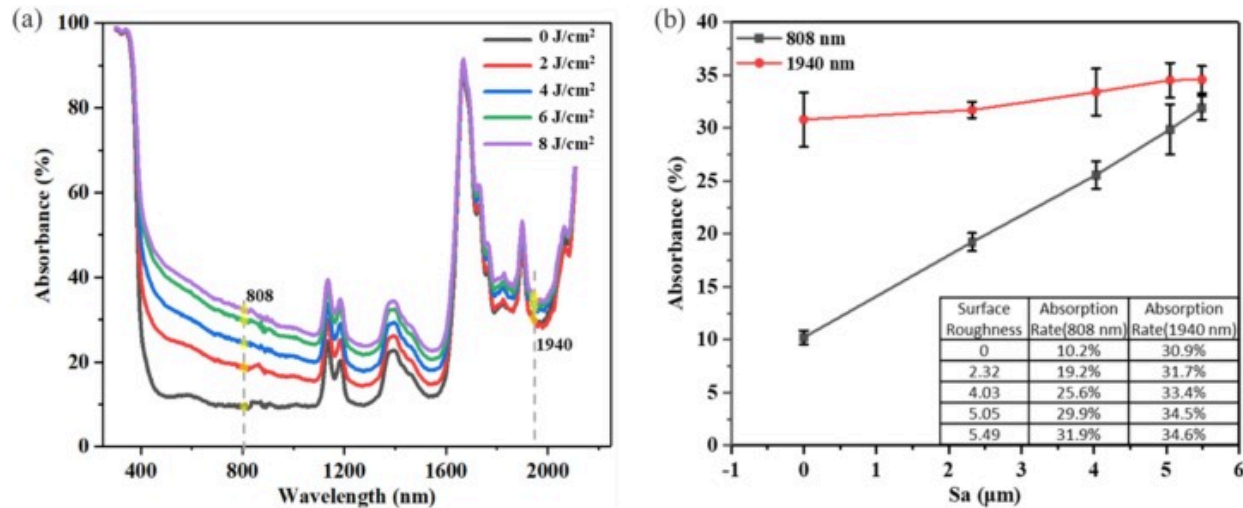
Download: [Download high-res image \(1MB\)](#)

Download: [Download full-size image](#)

Fig. 5. Physical appearance and LSCM images of PC samples after femtosecond laser ablation modification.

The absorption rates of the modified PC samples (exclusively modified by femtosecond laser ablation) for different laser wavelengths are shown in Fig. 6(a). Overall, as the energy density increased, the absorption rates of the samples for the studied laser wavelengths also increased to varying degrees. Since the welding lasers had wavelengths of 808nm and 1940nm, this study mainly examined the effect of surface roughness on the absorption rates of these two wavelengths, as shown in Fig. 6(b). With increasing roughness, the absorption rate of the PC samples for the 808nm laser increased significantly, reaching a maximum ($\sim 32.3\%$) when Sa was $5.49\mu\text{m}$. However, femtosecond laser modification did not significantly enhance the absorption of the samples for the

1940nm laser, indicating that the surface morphology changes induced by femtosecond laser ablation had little effect on the absorption rate for this wavelength. The intrinsic absorption of the polymer material dominated the absorption at 1940nm.



[Download: Download high-res image \(235KB\)](#)

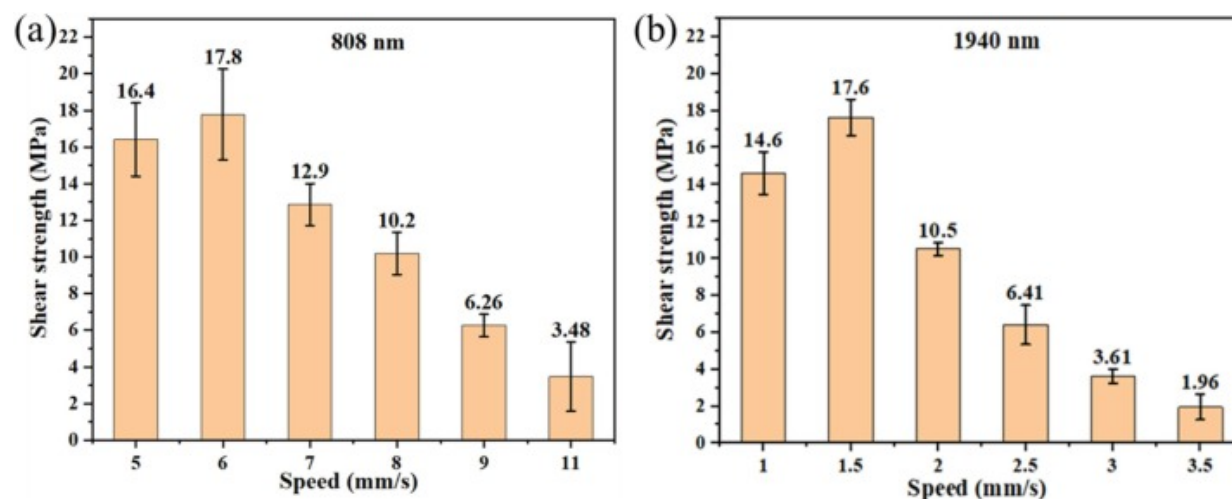
[Download: Download full-size image](#)

Fig. 6. (a) Absorption rates of PC samples after femtosecond laser ablation modification for different laser wavelengths;(b) Effect of surface roughness of PC samples on their laser absorption rate.

3.2. Welding results and analysis of PC/PVC

To compare the welding effects of 808nm and 1940nm lasers, experiments were conducted at the same welding power (11 W) while adjusting the welding speed to achieve optimal weld joints. In both laser wavelength experiments, PVC samples that had undergone femtosecond laser ablation modification (energy density: 8J/cm²) were used. The experimental results (Fig. 7) indicate that, within the investigated range of scanning speeds, the shear strength of the joints initially increased and then decreased as the welding speed increased, reaching peak values at 6mm/s and 1.5mm/s, respectively. Notably, the maximum joint strength for 808nm laser welding (17.8MPa) was close to

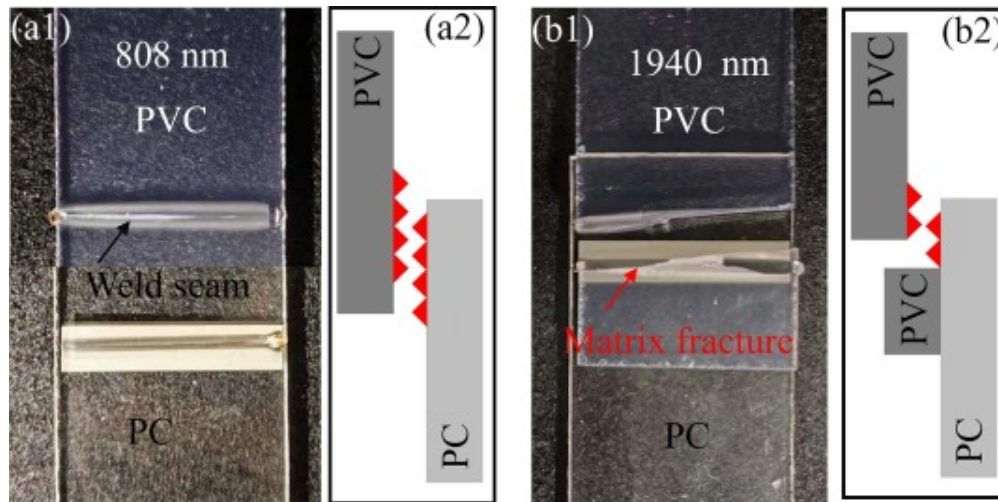
that of the 1940nm joint (17.6MPa), but the shear failure modes of the two were significantly different: the 808nm joints exhibited mixed failure (Fig. 8(a)), while the 1940nm joints experienced PVC matrix failure, with the PVC sample fracturing along the weld seam (Fig. 8(b)). Generally, the mixed failure mode indicates relatively insufficient bond strength at the interface, unable to fully resist external tensile stress. In contrast, the matrix failure mode indicates that the bond strength at the weld exceeds the tensile strength of the base material. However, in this experiment, the tensile strength of both laser-welded joints was almost identical. This difference may arise from the distinct welding processes of the two lasers: for the 808nm laser, due to the high transmittance of the polymer, only the weld seam area melts during welding, forming a “surface melt”; whereas the 1940nm laser causes almost the entire sample within the beam range to melt, resulting in a “bulk melt.” To gain a deeper understanding of the mechanisms behind NIR (808nm) and mid-infrared (1940nm) laser transmission welding, a multiphysics simulation analysis was performed for both welding processes.



[Download: Download high-res image \(150KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Effect of different scanning speeds on welding strength (11 W) (a) 808nm laser welding (b) 1940nm laser welding.



Download: [Download high-res image \(295KB\)](#)

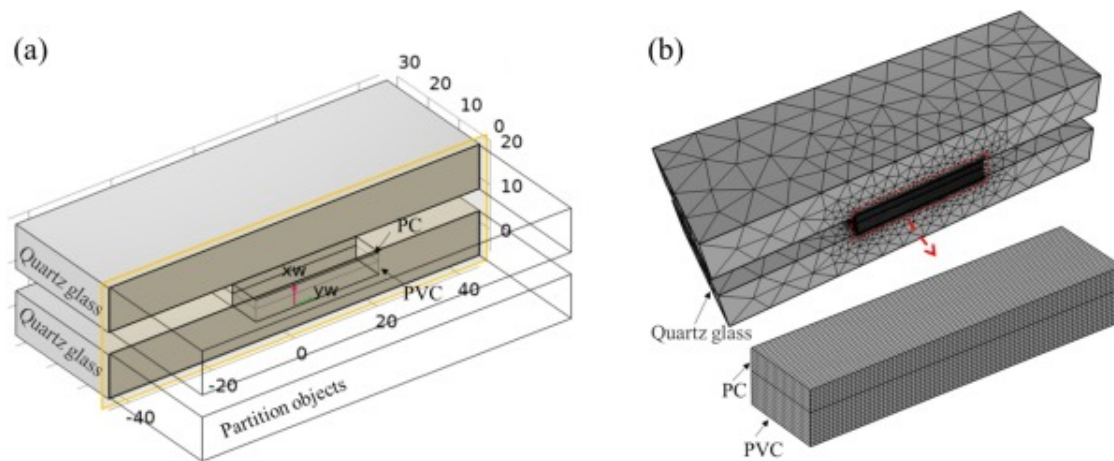
Download: [Download full-size image](#)

Fig. 8. Shear failure samples and failure mode diagram of laser welded joint: (a) Fracture of 808nm laser welded joint (11 W, 6mm/s, mixed failure); (b) Fracture of 1940nm laser welded joint (11 W, 1.5mm/s, PVC matrix failure).

3.3. Simulation of PC/PVC welding with 808nm and 1940nm lasers

To investigate the welding mechanisms of 808nm and 1940nm lasers, this study employs finite element analysis to numerically simulate the welding process. In this study, a 3D multiphysics model coupling heat transfer and solid mechanics was constructed using COMSOL to analyze the temperature field and residual stress distribution during the laser transmission welding of transparent PC/PVC. First, a 3D geometric model of the heat sink (quartz glass) and the samples (PC/PVC) was established, consisting of quartz glass with dimensions of $90 \times 60 \times 10 \text{ mm}^3$, PC of $30 \times 15 \times 2 \text{ mm}^3$, and PVC of $30 \times 15 \times 3 \text{ mm}^3$. The 3D model represents a typical symmetrical geometric structure, and to reduce computational load without affecting the analysis results, the model was halved along the laser movement direction using COMSOL's symmetry operation, as shown in Fig. 9(a). After creating the 3D models of the sample and the heat sink, the next step is to perform mesh

division. Mesh division is crucial as it defines the size and shape of the finite element sub-elements, which significantly impacting both the simulation results and computation speed. Theoretically, during the welding process, quartz glass undergoes minimal deformation and primarily serves as a heat conductor, so the mesh division for quartz glass uses the default tetrahedral mesh with standard accuracy. PC and PVC, being the main objects of analysis, experience significant temperature and stress variations during welding; therefore, the mesh division for these materials uses a swept hexahedral mesh with a maximum sub-element size of 0.3mm for higher accuracy, as shown in Fig. 9(b). The ambient temperature in the model is set to 20°C, and the outer surfaces of the quartz glass and samples (PC and PVC) are set with heat convection and radiation. The convective heat transfer coefficient is $10\text{W}/(\text{m}^2\cdot\text{K})$, and the emissivity is 0.92. The contact surfaces between the quartz glass, PC, and PVC are set to thermal conduction. Through the comparative analysis of coarse and fine models, as well as the convergence analysis of the simulation process, the model parameters were confirmed to be reasonable.



Download: [Download high-res image \(275KB\)](#)

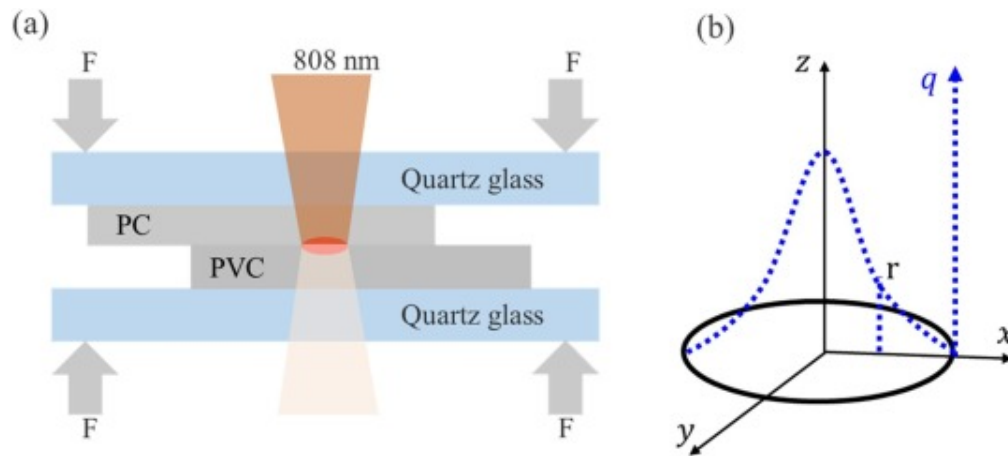
Download: [Download full-size image](#)

Fig. 9. (a) Geometric model of PC/PVC lap joint welding; (b) Schematic of COMSOL mesh division (Quartz: Tetrahedral mesh, Sample: Hexahedral mesh).

3.3.1. Laser beam and heat source

The welding heat source model forms the foundation for the numerical simulation of the welding process, and its selection directly impacts the accuracy of the finite element analysis results for the welding temperature field[29]. In LTW technology, surface heat source models, volumetric heat source models, or combined heat source models are typically used for numerical analysis depending on the characteristics of the heat source[17], [30], [31], [32], [33], [34], [35], [36], [37], [38]. The surface heat source model simulates the planar distribution of the heat source and is suitable for cases where the weld seam thickness is small and the molten pool is shallow, but it does not adequately capture the temperature distribution in the depth direction. The volumetric heat source model simulates the heat source distribution in the depth direction and is more appropriate for simulating welding processes with deeper molten pools. The combined heat source model uses multiple heat source models to simulate complex welding conditions, such as in hybrid welding technologies, but its parameter setup is challenging, and the computational load is substantial [27]. In the 808nm welding system, the cross-section of the laser beam after passing through the focusing lens forms a typical Gaussian beam. Since polymers have a very low absorption rate for 808nm lasers, in the model, the polymer's absorption rate for the laser is set to zero, considering only the absorption effect at the modified surface (PC/PVC interface). Therefore, in the 808nm laser welding model, it is more appropriate to define the heat source as a Gaussian surface heat source (Fig. 10), with the following equation.

$$H(r) = \frac{P}{\pi R^2} \exp\left(-\frac{r^2}{R^2}\right) \quad (2)$$



[Download: Download high-res image \(63KB\)](#)

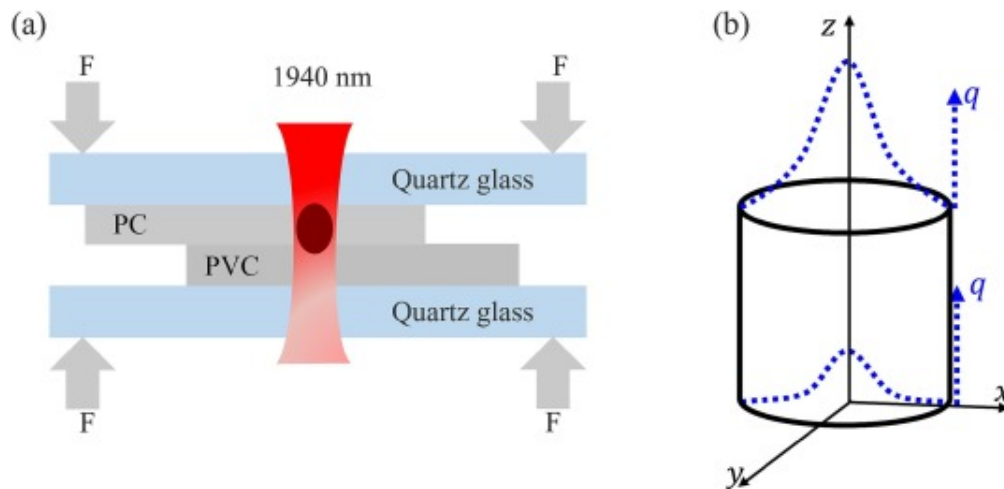
[Download: Download full-size image](#)

Fig. 10. Schematic of 808nm laser welding (surface heat source) (a) 808nm laser welding (surface absorption) (b) Energy density distribution of the Gaussian surface heat source beam cross-section.

In the equation, R is the spot radius, P represents the actual laser power absorbed by the material, and r denotes the distance from the point to the center of the spot. The power density at the center of the spot, $H(r = 0)$, is given by $P/\pi R^2$. The power density of the Gaussian surface heat source is determined by the input laser power and the size of the spot on the welding surface. For a given input power, the smaller the R , the higher the power density at the center.

The light source used in the mid-infrared laser welding experiments was a continuous thulium-doped fiber laser with a wavelength of 1940nm. The energy distribution of the laser beam cross-section emitted by the welding machine also follows a Gaussian profile. The Rayleigh length was measured to be approximately 50mm, which is significantly greater than the total thickness of the PC/PVC welding sample (5mm), indicating that the beam remains relatively focused throughout the welding process. For ease of calculation, in the numerical simulation analysis, the 1940nm laser beam passing through the

material was approximated as a collimated laser, with a Gaussian energy distribution across the cross-section, as shown in Fig. 11.



[Download: Download high-res image \(124KB\)](#)

[Download: Download full-size image](#)

Fig. 11. Schematic of 1940nm laser welding (volumetric heat source): (a) Schematic of 1940nm laser welding (volumetric absorption); (b) Schematic of energy density distribution in the cross-section of the Gaussian volumetric heat source beam.

The absorption of laser energy by polymers depends not only on the input laser power and spot radius but also on the polymer's intrinsic light absorption rate. Since polymers have a relatively high absorption rate for 1940nm lasers, the attenuation of laser power during propagation cannot be ignored. According to Beer-Lambert's law, and combined with **Eq. (2)**, the power density distribution of collimated light at various depths within the medium can be derived as shown in **Eq. (3)**. Based on the above analysis, this study chose to use the volumetric heat source model to accurately simulate energy absorption and heat transfer with depth [39], [40], [41]. Additionally, the experimental results have validated this choice, as they are in strong agreement with the predictions made using

the volumetric heat source model.

$$q(r, L) = \frac{P}{\pi R^2} \exp\left(\frac{-r^2}{R^2}\right) \exp(-\alpha L) = \frac{P}{\pi R^2} \exp\left(\frac{-r^2}{R^2} - \alpha L\right) \quad (3)$$

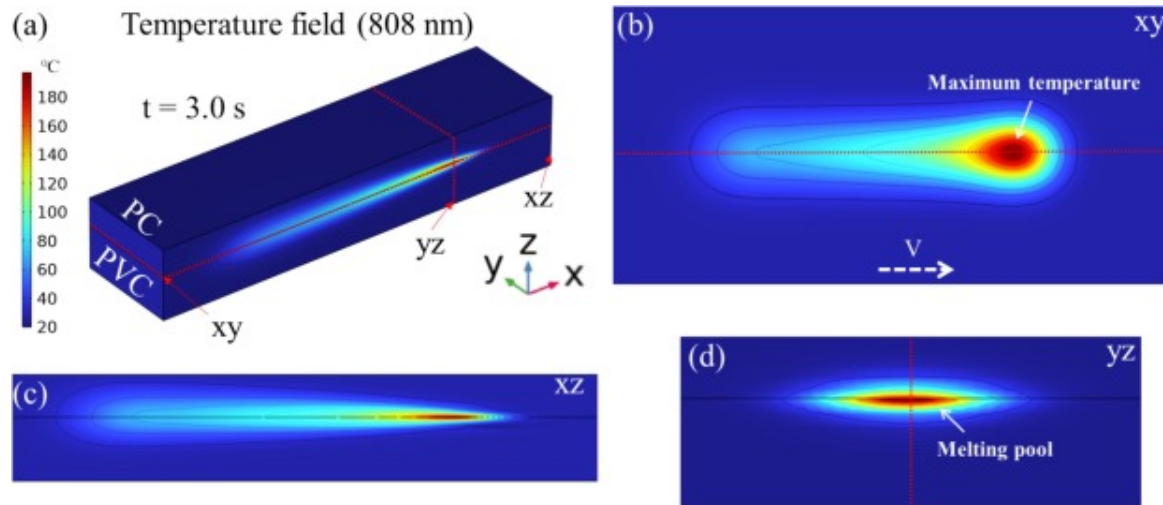
According to the differential form of Beer-Lambert's law (**Eq. 4**), the Gaussian volumetric heat source equation (**Eq. 5**) can be derived, where I represents the light intensity. In the simulation analysis, the material parameters built into the software were used, and coupling between the heat transfer and solid mechanics fields was achieved through thermal expansion.

$$\frac{\partial I}{\partial L} = \alpha * I \quad (4)$$

$$H(r, L) = \alpha * \frac{P}{\pi R^2} \exp\left(\frac{-r^2}{R^2} - \alpha L\right) \quad (5)$$

3.3.2. Simulation results and analysis

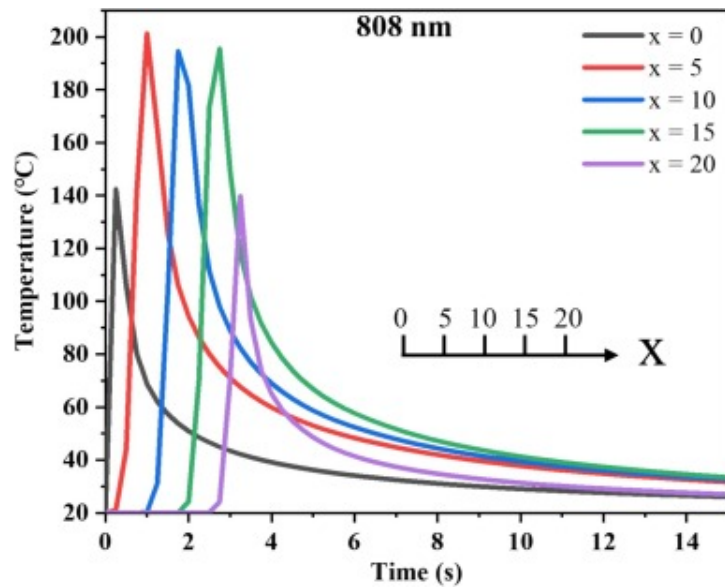
See [Fig. 12](#) shows the simulated temperature field contour plot for 808nm laser transmission welding of PC/PVC at a laser power of 11 W and a scanning speed of 6mm/s ($t=3$ s). It can be observed that the temperature field exhibits a comet-shaped morphology, with a narrower tail and the highest temperature located just behind the head of the comet. This is primarily due to the low thermal conductivity of polymers, resulting in a thermal lag, where the highest temperature is not at the center of the laser beam but slightly behind it. From the yz-plane cross-section of the weld ([Fig. 12 \(d\)](#)), the molten pool is centered at the welding interface and has an overall flattened shape. During welding, the laser passes through the upper PC layer and is directly absorbed by the modified area on the bottom surface of the PC. The laser energy is converted into heat energy (forming a surface heat source), causing the PC and PVC within the laser spot area to melt, thus achieving effective welding. [Fig. 13](#) shows the temperature curves at different positions along the center of the weld at various time points. During the welding process, the maximum temperature at the weld center is approximately 200°C.



[Download: Download high-res image \(195KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Temperature field contour plot of 808nm laser welding (a) isometric view (b) xy-plane cross-section (c) xz-plane cross-section (d) yz-plane cross-section.



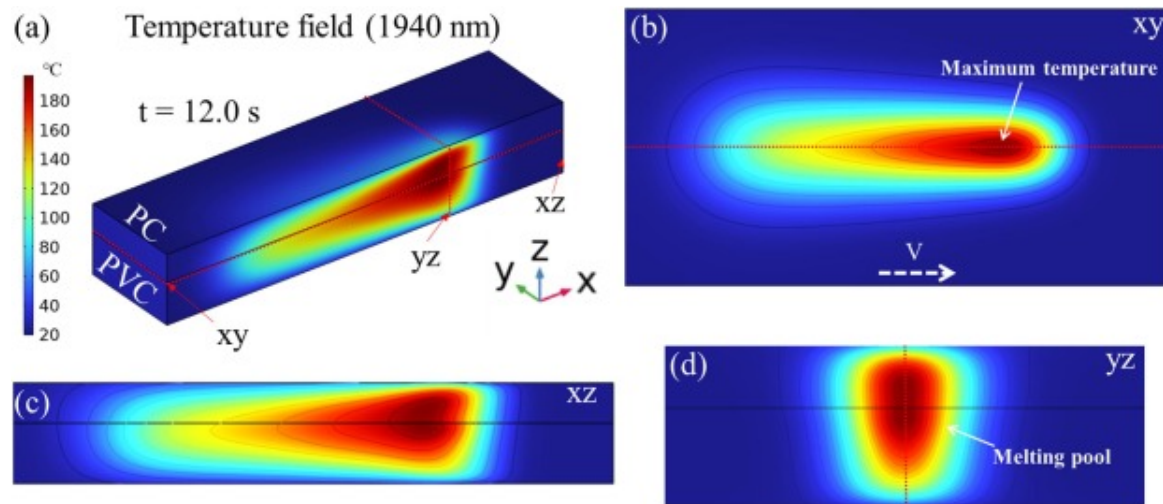
[Download: Download high-res image \(167KB\)](#)

[Download: Download full-size image](#)

Fig. 13. Temperature variation curves at different positions along the weld center during 808nm laser welding.

See Fig. 14 shows the simulated temperature field contour plot for 1940nm laser transmission welding of PC/PVC at a laser power of 11W and a scanning speed of 1.5 mm/s ($t=12$ s). The overall temperature field displays a comet-like shape, and the temperature field in the xy-plane (Fig. 14 (b)) is similar to the 808nm welding simulation. However, a significant difference is observed in the yz-plane cross-section, where the molten pool takes on an inverted cone shape, wider at the top and narrower at the bottom (Fig. 14 (d)). This is because polymers have a higher absorption rate for 1940nm laser, leading to more light energy absorption in the upper portion of the polymer, which gradually decreases toward the bottom. Notably, the top surface of the polymer is pressed against high thermal conductivity quartz glass, keeping the temperature at the glass contact surface relatively low. This prevents the polymer's upper surface from undergoing severe thermal degradation or carbonization during welding due to high temperatures.

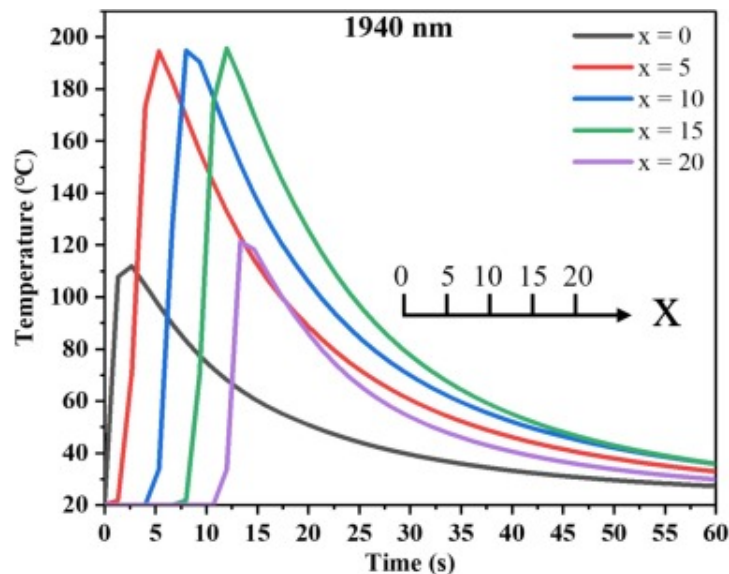
Fig. 15 presents the temperature curves along the weld center at various times, showing a maximum temperature of approximately 198°C . Furthermore, the broader heating range of the volumetric heat source leads to greater overall light energy absorption, resulting in a slower cooling rate compared to the 808 nm laser welding model.



Download: [Download high-res image \(206KB\)](#)

Download: [Download full-size image](#)

Fig. 14. Temperature field contour plot of 1940 nm laser welding (a) isometric view (b) xy-plane cross-section (c) xz-plane cross-section (d) yz-plane cross-section.



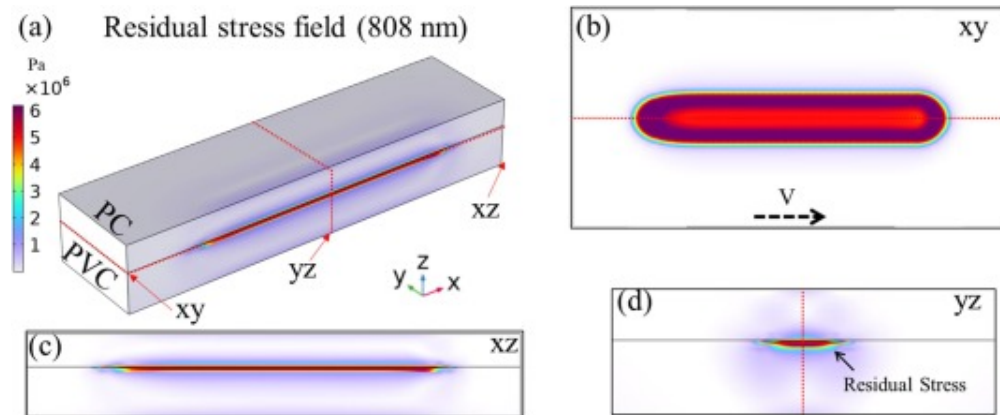
[Download: Download high-res image \(185KB\)](#)

[Download: Download full-size image](#)

Fig. 15. Temperature variation curves at different positions along the weld center during 1940 nm laser welding.

During the LTW process, the polymer locally absorbs laser energy, causing thermal expansion, which generates internal stress within the polymer and can even lead to plastic deformation. Due to uneven heating or rapid cooling, residual stress may remain inside the polymer after welding. The magnitude and distribution of residual stress significantly influence the service life and fracture strength of welded components. To analyze the residual stress, this study coupled the temperature and solid mechanics fields through thermal expansion, simulating the residual stress fields for 808nm and 1940nm laser welding. Fig. 16 shows the simulated residual stress field contour plot for 808nm laser welding of PC/PVC ($T < 25^{\circ}\text{C}$). The residual stress is mainly concentrated at the interface of the samples, and the stress zone exhibits an elliptical structure with a thickness of approximately 0.5mm. This occurs because the polymer in the molten pool and adjacent areas expands during heating, compressing the surrounding material and

causing plastic deformation. During cooling, the deformed polymer cannot fully return to its original position, leading to significant stress at the weld seam. As a result, the residual stress field following 808nm laser welding mirrors the shape of the molten pool.



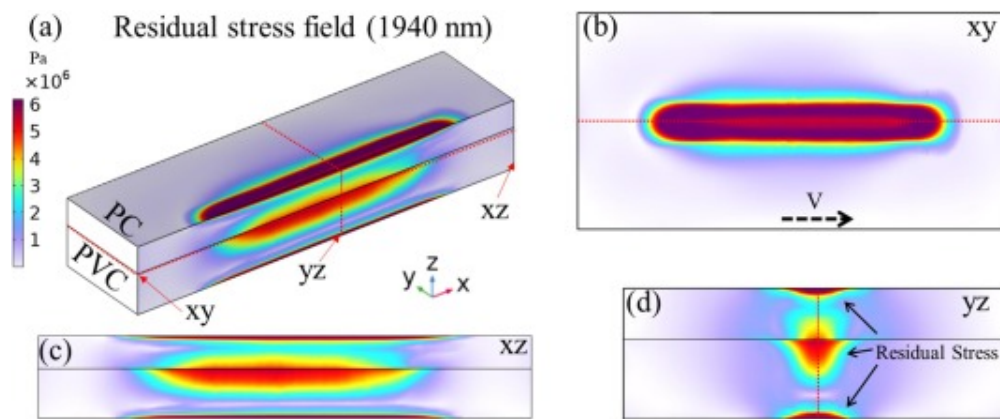
[Download: Download high-res image \(163KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Residual stress field contour plot after cooling in 808nm laser welding (a) isometric view (b) xy-plane cross-section (c) xz-plane cross-section (d) yz-plane cross-section.

See [Fig. 17](#) shows the simulated residual stress field contour plot for 1940nm laser welding of PC/PVC at temperatures below 25°C. Residual stress primarily accumulates in three areas: the upper surface of the PC, the lower surface of the PVC, and the PC/PVC interface. A toroidal molten pool formed near the PC/PVC interface ([Fig. 14\(d\)](#)) during welding. High temperatures within the pool led to material expansion, compressing the surrounding polymer and causing plastic deformation. The clamping pressure, combined with the high elastic modulus of the quartz glass, further intensified plastic deformation in the polymer at the contact interface. Additionally, the high thermal conductivity of the quartz glass caused a large temperature gradient near the polymer-glass contact surface, leading to the generation of residual stress on both the upper and lower surfaces of the polymer after cooling. A residual stress zone, narrower at the top and wider at the

bottom, was also observed near the PC/PVC interface (Fig. 17 (d)). This may be due to the slightly higher elastic modulus of PC and its lower coefficient of thermal expansion compared to PVC, resulting in smaller deformation during the welding process and, consequently, a smaller residual stress region. A comparison between the simulation results of 808nm and 1940nm laser welding shows that residual stress for the 808nm process is mainly concentrated around the weld seam. In contrast, 1940nm laser welding induces residual stress not only near the weld seam but also at the polymer-quartz glass contact interface. Additionally, the residual stress on the PVC side near the weld seam covers a broader area. Excessive residual stress peaks or ranges can reduce the material's mechanical strength and may lead to warping, cracking, or other phenomena, further affecting the mechanical strength of the sample. This is the primary reason for the PVC matrix fracture observed in the 1940nm laser-welded joints.



[Download: Download high-res image \(191KB\)](#)

[Download: Download full-size image](#)

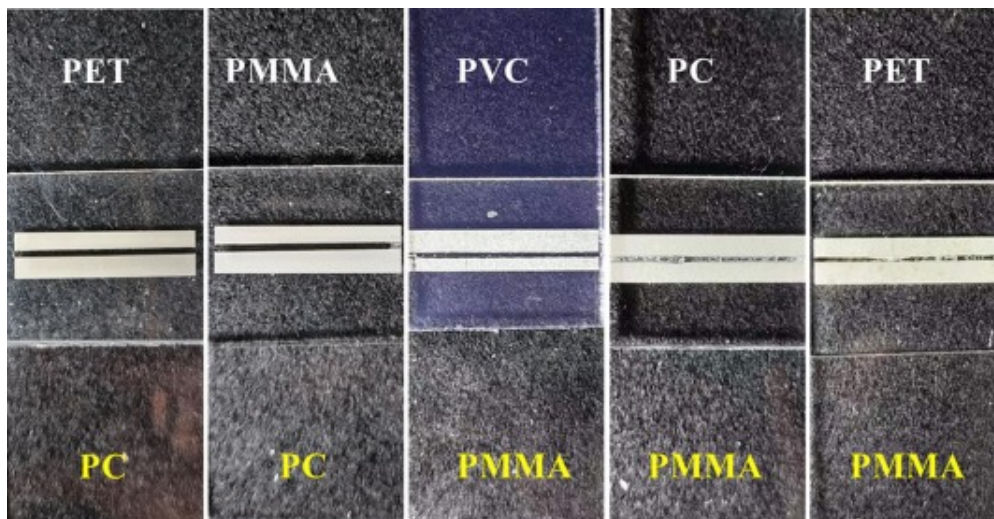
Fig. 17. Residual stress field contour plot after cooling in 1940 nm laser welding (a) isometric view (b) xy-plane cross-section (c) xz-plane cross-section (d) yz-plane cross-section.

The analysis suggests that the 1940nm mid-infrared laser welding alters the inherent properties of the base material, introducing residual stress and reducing its mechanical

strength. This makes it suitable for applications involving thin materials with inherently high strength. On the other hand, the 808nm NIR laser welding, which causes minimal alteration to the base material, is better suited for thicker materials or scenarios where maintaining the integrity of the base material is critical.

3.4. Validation on other polymer combinations

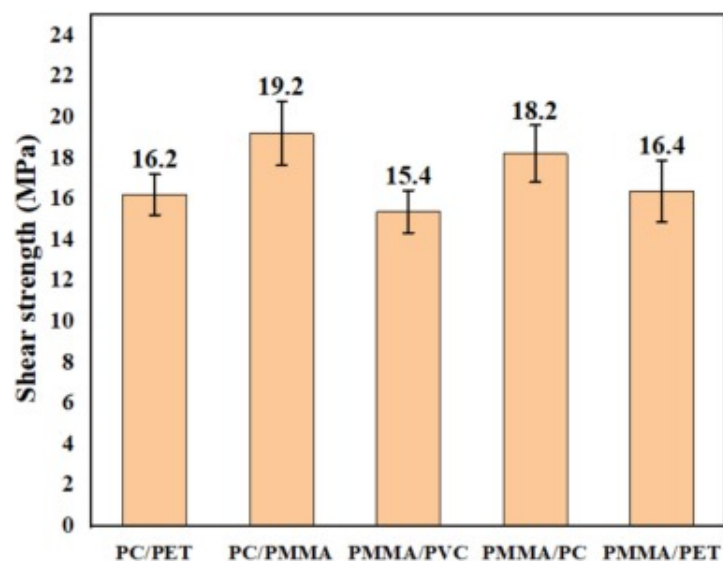
To validate the applicability of the femtosecond laser ablation method on different polymers, this study first applied femtosecond laser ablation to PC and PMMA, followed by 808nm NIR laser transmission welding on the following combinations: PC (ablated)/PMMA, PC (ablated)/PET, PMMA (ablated)/PVC, PMMA (ablated)/PC, and PMMA (ablated)/PET. The welded samples are shown in [Fig. 18](#). All samples were processed under consistent modification parameters ($8\text{J}/\text{cm}^2$) and welding parameters (11W, 6mm/s). The shear strength test results of the welded samples, as shown in [Fig. 19](#), indicate that the shear strength of all samples exceeded 15MPa.



[Download: Download high-res image \(323KB\)](#)

[Download: Download full-size image](#)

Fig. 18. Samples of NIR laser (808nm) transmission welding for different transparent polymers.



[Download: Download high-res image \(105KB\)](#)

[Download: Download full-size image](#)

Fig. 19. Shear strength of welded joints for different transparent polymers.

4. Conclusions

To achieve NIR laser welding of two transparent polymers without absorbers, this paper proposes a method of using femtosecond laser ablation on the polymer surface prior to welding to enhance its absorption of NIR laser energy, and the feasibility of this method was validated. Initially, GFL was applied to ablate the surface of the PC sample, followed by measurements of surface roughness and absorption rate. Subsequently, transmission welding of PC/PVC was performed using 808nm and 1940nm lasers, without absorbers. Additionally, the shear strength of the welded joints was examined, and their failure mechanisms were analyzed. Multiphysics simulations were conducted for both welding

processes to aid in identifying the reasons behind the differing joint failure modes.

Finally, the applicability of the femtosecond laser ablation method for NIR laser welding without absorbers was validated across multiple sets of transparent compatible polymers. This approach enhances the safety, environmental friendliness, and cost-effectiveness of LTW technology, making it highly promising for applications in the biomedical, food, and medical fields. The main conclusions of this paper are summarized as follows:

1. Femtosecond laser ablation was proposed and experimentally validated as a method to enhance the absorption rate of polymers for NIR laser energy before welding. Transmission welding of modified PC (exclusively modified by femtosecond laser ablation) and PVC was achieved using an 808nm NIR laser without absorbers.
2. Femtosecond laser ablation of the surface of transparent polymers can enhance their absorption capacity for 808nm NIR laser energy. However, femtosecond laser ablation does not significantly improve the absorption rate of polymers for 1940nm mid-infrared laser energy.
3. Mechanical tests on the welded joints of 808nm and 1940nm lasers revealed that the maximum shear strength of both joints is similar, at 17.8MPa and 17.6MPa, respectively. However, the failure mode of the former is mixed failure, while the latter experiences PVC matrix failure.
4. Multiphysics simulations of the 808nm and 1940nm laser welding processes revealed that residual stress in the 808nm joints primarily concentrates near the weld seam. In contrast, residual stress in the 1940nm joints is observed both at the upper and lower surfaces (in contact with quartz) and at the weld seam, leading to a higher likelihood of PVC matrix fracture.
5. To verify the applicability of this method, femtosecond laser ablation was applied to PC and PMMA, followed by 808nm NIR laser transmission

welding of multiple sets of transparent compatible polymers (PC/PMMA, PC/PET, PMMA/PVC, PMMA/PC, and PMMA/PET) without absorbers. All samples achieved shear strengths greater than 15MPa.

Funding.

Shenzhen Technology University Enterprise Cooperation Research Fund (2018010802013); National Key Research and Development Program of China (2022YFF0710800); National Key Research and Development Program of China (2022YFF0710802); National Natural Science Foundation of China (62071311).

CRediT authorship contribution statement

Dongyang Wang: Writing – review & editing, Writing – original draft, Software, Investigation, Data curation, Conceptualization. **Kaige Liu:** Methodology, Investigation, Formal analysis. **Jianpang Zhai:** Visualization, Validation, Supervision, Conceptualization. **Shuangchen Ruan:** Visualization, Validation, Supervision. **Yan Kang:** Visualization, Supervision, Resources, Project administration, Funding acquisition. **Qingyang Wu:** Visualization, Validation, Supervision, Software, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] B. Acherjee
Laser transmission welding of polymers-a review on process fundamentals, material attributes, weldability, and welding techniques
J MANUF PROCESS, 60 (2020), pp. 227-246
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] M. Mehrpouya, A. Gisario, A. Rahimzadeh, M. Barletta
An artificial neural network model for laser transmission welding of biodegradable polyethylene terephthalate/polyethylene vinyl acetate (PET/PEVA) blends
Int. J. Adv. Manuf. Technol., 102 (2019), pp. 1497-1507
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] J. Jiao, J. Xu, C. Jing, L. Sheng, H. Ru, H. Xia
Laser welding process and strength enhancement of carbon fiber reinforced thermoplastic composites and metals dissimilar joint: A review
CHINESE J AERONAUT, 36 (2023), pp. 13-31
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] B. Acherjee
State-of-art review of laser irradiation strategies applied to laser transmission welding of polymers
OPT LASER TECHNOL, 137 (2021), Article 106737
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] B. Acherjee
Laser transmission welding of polymers-a review on welding parameters, quality attributes, process monitoring, and applications
J MANUF PROCESS, 64 (2021), pp. 421-443
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] H. Wan, J. Min, J. Lin

Experimental and theoretical studies on laser treatment strategies for improving shear bonding strength of structural adhesive joints with cast aluminum

COMPOS STRUCT, 279 (2022), Article 114831

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[7]

D. Kumar Goyal, R. Yadav, R. Kant

Laser transmission welding of polycarbonate sheets using electrolytic iron powder absorber

OPT LASER TECHNOL, 161 (2023), Article 109165

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[8]

X. Yu, H. Qiao, Z. Chen, Y. Chen, Q. Li, Y. Liu, C. Wang

Mechanism and factor analysis of welding PC and PASF using magnesium zinc alloy particles as laser-absorbent

Opt. Laser Technol., 158 (2023), Article 108859

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[9]

A. Olowinsky, and A. Boglea, "Extending the process limits of laser polymer welding with high-brilliance beam sources (recent status and prospects of polybright)," (SPIE, Bellingham, Wash, 2011), pp. 792107-7921015.

[Google Scholar ↗](#)

[10]

Y. Chen, H. Qiao, Q. Li, X. Yu, C. Wang

Influences of laser profiles on the weld quality of glass fiber reinforced PBT using laser transmission welding

OPT LASER TECHNOL, 174 (2024), Article 110672

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[11]

U.A. Russek, A. Palmen, H. Staub, J. Poehler, C. Wenzlau, G. Otto, M. Poggel, A. Koeppe, H. Kind

Laser beam welding of thermoplastics

In *Photon Processing in Microelectronics and Photonics II*(SPIE (2003), pp. 458-472

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [12] F. Becker, H. Potente
A step towards understanding the heating phase of laser transmission welding in polymers
Polym. Eng. Sci., 42 (2002), pp. 365-374
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] P.A. Hilton, I.A. Jones, Y. Kennish
Transmission laser welding of plastics
In *First International Symposium on High-Power Laser Macroprocessing*(SPIE (2003), pp. 44-52
[Google Scholar ↗](#)
- [14] X. Jiang, S. Chandrasekar, C. Wang
A laser microwelding method for assembly of polymer based microfluidic devices
OPT LASER ENG, 66 (2015), pp. 98-104
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [15] M. Liu, D. Ouyang, J. Zhao, C. Li, H. Sun, S. Ruan
Clear plastic transmission laser welding using a metal absorber
OPT LASER TECHNOL, 105 (2018), pp. 242-248
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [16] M. Liu, D. Ouyang, C. Li, H. Sun, S. Ruan
Effects of metal absorber thermal conductivity on clear plastic laser transmission welding
CHINESE PHYS LETT, 35 (2018), Article 104205
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] C. Wang, X. Yu, M. Jiang
Fracture forms and multiple undulated morphology of polycarbonate LTW by introducing carbon black coated copper film
Opt. Laser Technol., 141 (2021), Article 107132
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [18] Z. Chen, Z. Wu, S. Lu, G. Zhang, H. Yan
A new laser welding method of two transparent plastic parts based on reflector assisted method
J. Mater. Res. Technol., 24 (2023), pp. 5309-5320
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [19] M. Aden, V. Mamuschkin, A. Olowinsky
Influence of carbon black and indium tin oxide absorber particles on laser transmission welding
OPT LASER TECHNOL, 69 (2015), pp. 87-91
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [20] N. M. Woosman, and L. P. Frieder, "Clearweld: welding of clear, coloured, or opaque thermoplastics," Proceedings of the Institution of Mechanical Engineers. Part D, Journal of Automobile Engineering **219**, 1069-1074 (2005).
[Google Scholar ↗](#)
- [21] X. Yu, C. Ni, H. Qiao, Y. Chen, Y. Liu, C. Wang
The effects of laser line energy on welding strength of PASF/PC using magnesium zinc alloy powders as absorbents in laser transmission welding
Opt. Laser Technol., 174 (2024), Article 110702
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] I. Mingareev, F. Weirauch, A. Olowinsky, L. Shah, P. Kadwani, M. Richardson

Welding of polymers using a 2 microns thulium fiber laser

OPT LASER TECHNOL, 44 (2012), pp. 2095-2099

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[23] V. Mamuschkin, C. Engelmann, A. Olowinsky

Improvement of energy deposition in absorber-free laser welding through quasi-simultaneous irradiation

Phys. Procedia, 83 (2016), pp. 472-482

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[24] M. Devrient, T. Frick, M. Schmidt

Laser transmission welding of optical transparent thermoplastics

Phys. Procedia, 12 (2011), pp. 157-165

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[25] J. De Pelsmaeker, G. Graulus, S. Van Vlierberghe, H. Thienpont, D. Van Hemelrijck, P. Dubruel, H. Ottevaere

Clear to clear laser welding for joining thermoplastic polymers: a comparative study based on physicochemical characterization

J MATER PROCESS TECH, 255 (2018), pp. 808-815

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[26] G. Yujun, L. Changsheng, H. Chunlin, C. Qingkui, and Z. Xiaowen, "89-1 coating for laser surface hardening," Laser Technology (2002).

[Google Scholar ↗](#)

[27] D. Wang, D. Ouyang, Y. Chen, S. Xu, Q. Lv, J. Zhai, M. Liu, S. Ruan

Laser transmission welding of glass-fiber-reinforced polymer and stainless steel with a three-dimensional-printing surface pretreatment

Opt. Laser Technol., 168 (2024), Article 109967

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [28] D. Wang, M. Liu, S. Xu, J. Lu, Y. Zhang, J. Zhai, S. Ruan, D. Ouyang, C. Li
Laser transmission welding of Poly(vinyl chloride) and polyamide 66
based on green femtosecond laser surface modification
Opt. Laser Technol., 181 (2025), Article 111737
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] S. Hu, F. Li, P. Zuo
Numerical simulation of laser transmission welding-a review on
temperature field, stress field, melt flow field, and thermal degradation
POLYMERS-BASEL, 15 (2023), p. 2125
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] B. Acherjee, A.S. Kuar, S. Mitra, D. Misra
Effect of carbon black on temperature field and weld profile during laser
transmission welding of polymers: a FEM study
OPT LASER TECHNOL, 44 (2012), pp. 514-521
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] Y. Ai, K. Zheng, Y.C. Shin, B. Wu
Analysis of weld geometry and liquid flow in laser transmission welding
between polyethylene terephthalate (PET) and Ti6Al4V based on
numerical simulation
Opt. Laser Technol., 103 (2018), pp. 99-108
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] D. Kumar, N.S. Sarkar, B. Acherjee, A.S. Kuar
Beam wobbling effects on laser transmission welding of dissimilar
polymers: experiments, modeling, and process optimization
Opt. Laser Technol., 146 (2022), Article 107603
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] B. Acherjee, A.S. Kuar, S. Mitra, D. Misra

Modeling of laser transmission contour welding process using FEA and DoE

Opt. Laser Technol., 44 (2012), pp. 1281-1289



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [34] C. Wang, X. Yu, M. Jiang, Z. Xing, C. Wang
Numerical and experimental investigation into the evolution and distribution of residual stress in laser transmission welding of PC/Cu/PC

Opt. Laser Technol., 136 (2021), Article 106786



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [35] B. Acherjee
Numerical study of thermo-mechanical responses in laser transmission welding of polymers using a 3-D thermo-elasto-viscoplastic FE model

WELD WORLD, 66 (2022), pp. 1421-1435

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [36] C. Wang, H. Liu, Z. Chen, D. Zhao, C. Wang
A new finite element model accounting for thermal contact conductance in laser transmission welding of thermoplastics

INFRARED PHYS TECHN, 112 (2021), Article 103598



[View PDF](#) [View article](#) [Google Scholar ↗](#)

- [37] H. Liu, W. Liu, D. Meng, X. Wang
Simulation and experimental study of laser transmission welding considering the influence of interfacial contact status

MATER DESIGN, 92 (2016), pp. 246-260



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [38] N. Nguyen, S. Behrens, M. Brosda, A. Olowinsky, A. Gillner

Modelling and thermal simulation of absorber-free quasi-simultaneous laser welding of transparent plastics

WELD WORLD, 64 (2020), pp. 1939-1946

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [39] B. Acherjee, A.S. Kuar, S. Mitra, D. Misra
Modeling and analysis of simultaneous laser transmission welding of polycarbonates using an FEM and RSM combined approach

Opt. Laser Technol., 44 (2012), pp. 995-1006



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [40] A. Duggirala, P. Kalvettukaran, B. Acherjee, S. Mitra
Numerical simulation of the temperature field, weld profile, and weld pool dynamics in laser welding of aluminium alloy

Optik (stuttgart), 247 (2021), Article 167990



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [41] B. Acherjee
3-D FE heat transfer simulation of quasi-simultaneous laser transmission welding of thermoplastics

J BRAZ SOC MECH SCI, 41 (2019), pp. 1-13

[Google Scholar ↗](#)

[View Abstract](#)

Recommended articles

Based on reading popularity

[Research article](#) [Full text access](#)

Fs-laser based hybrid micromachining for polymer micro-opto electrical systems

Gian-Luca Roth, ..., Ralf Hellmann

Optics and Lasers in Engineering • Volume 137 • 2021 • Article 106362



[View PDF](#)

Research article Full text access

On enhancement of fracture resistance of adhesive joints by surface micropatterning using a femtosecond laser

Fengzhen Sun, ..., Bamber R.K. Blackman

Journal of Materials Processing Technology • Volume 315 • 2023 • Article 117904



[View PDF](#)

Research article Open access

Influence of laser texturing strategy on thermomechanical joining of AA7075 aluminum alloy and PEEK

Francesco Lambiase, ..., Alfonso Paoletti

Composite Structures • Volume 315 • 2023 • Article 116974



[View PDF](#)

Review article Open access

Advances in reaction interface engineering for 3D hydrogel and aerogel derived electrodes in energy conversion devices

Jiayi Tang, ..., Zongping Shao

Energy Reviews • Volume 4, Issue 3 • 2025 • Article 100153



[View PDF](#)

Research article Full text access

Numerical simulation of thermal distribution and residual stress characteristic for laser wobble joining of CFRTP and Ti-6Al-4V alloy

Hengchang Bu, ..., Wanping Ma

Journal of Manufacturing Processes • Volume 79 • 2022 • Pages 562-575



[View PDF](#)

Research article Open access

Teaching advanced manufacturing through course-based undergraduate research in a vertically integrated projects (VIP) class

Harry Watkins, Sourabh K. Saha

Manufacturing Letters • Volume 44, Supplement • 2025 • Pages 1628-1636



[View PDF](#)

Cited by (6)

Research article

A comparative study on laser shock deformation of silver nanowire junctions with different sizes for transparent conductors

Hu Y., ..., Hu Y.

Optics and Laser Technology • Volume 199 • 2026 • Article 115005

[Show abstract](#) ✓

Research article

Breaking thickness-strength limitations in plastic welding via a dual-wavelength laser strategy with deep penetration and extended process window

Chen Y., ..., Wang C.

Optics and Laser Technology • Volume 193 • 2026 • Article 114171

[Show abstract](#) ✓

Research article

Enhancing the welding joint strength of aluminum nitride ceramic and polycarbonate by micro-texture treatment

Jia F., ..., Wang C.

Ceramics International • Volume 51 • 2025

[Show abstract](#) ✓

Research on Process Optimization of Laser Transmission Welding of PC Materials Based on Bezier Curves ↗

Yan Y., ..., Wang C.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

Metrics

Citations

Citation Indexes	6
------------------	---

Captures

Mendeley Readers	11
------------------	----



[View details ↗](#)

© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.